

- 1) Plot the given complex number and also find the modulus and argument of z where $z = -1 + i\sqrt{3}$

Solution:

Given, $z = -1 + i\sqrt{3}$

$$|z| = \sqrt{x^2 + y^2}$$

$$= \sqrt{(-1)^2 + (\sqrt{3})^2}$$

$$= 2$$

$$\therefore \text{Arg}(z) = \tan^{-1}\left(\frac{y}{x}\right)$$

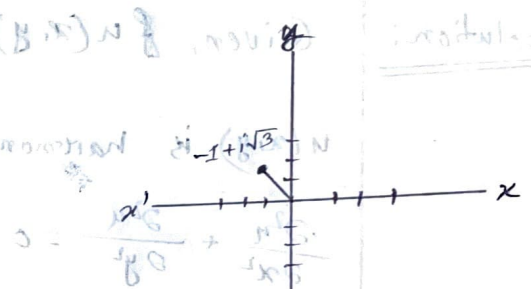
$$= \tan^{-1} \frac{\sqrt{3}}{-1}$$

$$= \tan^{-1} \sqrt{3}$$

$$= \pi - \frac{\pi}{3}$$

$$= \frac{3\pi - \pi}{3}$$

$$= \frac{2\pi}{3}$$



- 2) Show that the function $u(x, y) = x^2 - y^2 - 2y$ is a Harmonic function. Find its Harmonic Conjugate $v(x, y)$ and construct the corresponding Analytic function $f(z) = u + iv$.

Solution:

Given, $u(x, y) = x^2 - y^2 - 2y$

$u(x, y)$ is harmonic function if it satisfies Laplace's equation:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\therefore \frac{\partial u}{\partial x} = 2x \quad \therefore \frac{\partial u}{\partial y} = -2y - 2$$

$$\therefore \frac{\partial^2 u}{\partial x^2} = 2 \quad \therefore \frac{\partial^2 u}{\partial y^2} = -2$$

$$\therefore \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0$$

$\therefore u(x, y)$ is a Harmonic function.

Now, The harmonic conjugate v must satisfy the Cauchy-Riemann equations:

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ --- (1) and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \text{ --- (2)}$$

From equation (1) we get,

$$\frac{\partial v}{\partial y} = 2x \text{ --- (3)}$$

From equation ② we get:

$$\frac{\partial v}{\partial x} = 2y + 2 \quad \text{--- ④}$$

From equation ③ we get,

$$\frac{\partial v}{\partial y} = 2x$$

$$\Rightarrow \int \partial v = \int 2x \partial y$$

$$\Rightarrow v = 2xy + F(x) \quad \text{--- ⑤} \quad [\text{Where } F(x) \text{ is a function of } x]$$

Now, partially differentiate with respect to x ,

$$\frac{\partial v}{\partial x} = \frac{\partial}{\partial x} (2xy + F(x))$$

$$\Rightarrow \cancel{2y} + 2 = 2y + F'(x) \quad \left[\text{From ④, } \frac{\partial v}{\partial x} = 2y + 2 \right]$$

$$\Rightarrow F'(x) = 2$$

$$\Rightarrow \int F'(x) dx = \int 2 dx$$

$$\Rightarrow F(x) = 2x + C$$

Putting the value of $F(x)$ into ⑤,

$$v = 2xy + 2x + C$$

$$\therefore f(z) = u + iv = x^2 - y^2 - 2y + i(2xy + 2x + C)$$

$$= x^2 - y^2 - 2y + 2ixy + 2ix + iC = (x^2 - y^2 + 2ixy) + 2i(x + y) + iC$$

$$= z^2 + 2iz + iC$$

3)

Evaluate $\oint_C \frac{z+2}{z(z-1)} dz$ where C is $|z|=1$.Solution:

$$\text{Let, } \frac{z+2}{z(z-1)} = \frac{A}{z} + \frac{B}{z-1}$$

$$\Rightarrow z+2 = A(z-1) + Bz$$

$$\Rightarrow z+2 = Az - A + Bz$$

$$\Rightarrow z+2 = (A+B)z - A$$

By Comparing coefficients, $A+B=1$ and $A=-2$

$$\therefore -2+B=1$$

$$\therefore B=3$$

Thus,

$$\frac{z+2}{z(z-1)} = \frac{-2}{z} + \frac{3}{z-1}$$

$$\therefore \oint_C \frac{z+2}{z(z-1)} dz = \oint_C \left(1 + \frac{2}{z-1}\right) dz = \oint_C 1 dz + \oint_C \frac{2}{z-1} dz$$

Applying Cauchy's Integral Theorem,

$$\oint_C 1 dz = 0$$

Applying Cauchy's Integral Formula,

$$\oint_C \frac{2}{z-1} dz = 2 \oint_C \frac{1}{z-1} dz = 2 \times 2\pi i = 4\pi i$$

$$\therefore \oint_C \frac{z+2}{z(z-1)} dz = 0 + 4\pi i = 4\pi i$$

4) a) Expand $f(z) = \frac{1}{(z-1)(z-4)}$ in a Laurent's series valid for $1 < |z| < 4$

Solution:

$$\text{Let, } \frac{1}{(z-1)(z-4)} = \frac{A}{z-1} + \frac{B}{z-4}$$

$$\Rightarrow A(z-4) + B(z-1) = 1$$

$$\Rightarrow Az - 4A + Bz - B = 1$$

$$\Rightarrow z(A+B) - (4A+B) = 1$$

$\Rightarrow$

Comparing coefficients,

$$A+B=0 \Rightarrow A=-B$$

$$\text{and, } -4A-B=1$$

$$\Rightarrow -4A+A=1$$

$$\Rightarrow -3A=1$$

$$\therefore A = -\frac{1}{3}$$

$$\therefore B = \frac{1}{3}$$

$$\text{So, } f(z) = \frac{-1}{3(z-1)} + \frac{1}{3(z-4)} \quad \text{--- (1)}$$

$$\text{Since, } |z| > 1$$

$$\Rightarrow \left| \frac{1}{z} \right| < 1$$

$$\text{and } |z| < 4$$

$$\Rightarrow \left| \frac{z}{4} \right| < 1$$

$$\text{So, } f(z) = \frac{-1}{3(z-1)} + \frac{1}{3(z-4)}$$

$$= \frac{-1}{3z(1-\frac{1}{z})} + \frac{1}{3 \times 4(1-\frac{z}{4})}$$

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Since,

$$|z| > 1$$

$$\Rightarrow \left| \frac{1}{z} \right| < 1$$

and $|z| < 4$

~~$\left| \frac{z}{4} \right| < 1$~~
 $\Rightarrow \left| \frac{z}{4} \right| < 1$

From (1),

$$\begin{aligned} f(z) &= -\frac{1}{3z(1-\frac{1}{z})} - \frac{1}{12(1-\frac{z}{4})} \\ &= -\frac{1}{3} \left\{ \frac{1}{z(1-\frac{1}{z})} \right\} + \frac{1}{3} \left\{ -\frac{1}{4} \left(1 + \frac{z}{4} + \frac{z^2}{4^2} + \frac{z^3}{4^3} + \dots \right) \right\} \\ &= -\frac{1}{3} (z-1)^{-1} + \frac{1}{3} (z-4)^{-1} \\ &= -\frac{1}{3} \left(\frac{1}{z} + \frac{1}{z^2} + \frac{1}{z^3} + \dots \right) + \frac{1}{3} \left\{ -\frac{1}{4} \left(1 + \frac{z}{4} + \frac{z^2}{4^2} + \frac{z^3}{4^3} + \dots \right) \right\} \end{aligned}$$

$$\therefore \frac{1}{(z-1)(z-4)} = -\frac{1}{3z} - \frac{1}{3z^2} - \frac{1}{3z^3} + \dots + \left(-\frac{1}{12} - \frac{1}{48}z - \frac{1}{192}z^2 + \dots \right)$$

which is the required Laurent's series for $f(z)$ valid
for $1 < |z| < 4$

4) b) Expand $f(z) = \sin z$ in a Taylor series about $z = \frac{\pi}{3}$

Solution: Given that, $f(z) = \sin z$

$$\therefore f\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$f'(z) = \cos z$$

$$\therefore f'\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

$$f''(z) = -\sin z$$

$$\therefore f''\left(\frac{\pi}{3}\right) = -\frac{\sqrt{3}}{2}$$

$$f'''(z) = -\cos z$$

$$\therefore f'''\left(\frac{\pi}{3}\right) = -\frac{1}{2}$$

$\therefore$ Taylor series for $f(z) = \sin z$ about $z = \frac{\pi}{3}$ is given by:

$$f(z) = f\left(\frac{\pi}{3}\right) + \frac{f'\left(\frac{\pi}{3}\right)}{1!} \left(z - \frac{\pi}{3}\right) + \frac{f''\left(\frac{\pi}{3}\right)}{2!} \left(z - \frac{\pi}{3}\right)^2 + \frac{f'''\left(\frac{\pi}{3}\right)}{3!} \left(z - \frac{\pi}{3}\right)^3 + \dots$$

$$\Rightarrow \sin z = \frac{\sqrt{3}}{2} + \frac{\frac{1}{2}}{1!} \left(z - \frac{\pi}{3}\right) + \frac{\left(-\frac{\sqrt{3}}{2}\right)}{2!} \left(z - \frac{\pi}{3}\right)^2 + \frac{\left(-\frac{1}{2}\right)}{3!} \left(z - \frac{\pi}{3}\right)^3 + \dots$$

Which is the required Taylor series for $f(z)$

5)

Use the Cauchy's residue theorem to evaluate $\int_0^{2\pi} \frac{d\theta}{5+2\cos\theta}$ Solution:

$$\text{Let, } I = \int_0^{2\pi} \frac{d\theta}{5+2\cos\theta} \quad \text{--- (1)}$$

- Also, consider the unit circle $C: |z|=1$ for the present problem.

$$\text{Now, } |z|=1$$

$$\Rightarrow z = e^{i\theta} = \cos\theta + i\sin\theta \quad \text{--- (2)}$$

$$\therefore z^{-1} = \frac{1}{z} = e^{-i\theta} = \cos\theta - i\sin\theta \quad \text{--- (3)}$$

$$\text{(2) + (3)} \Rightarrow$$

$$2\cos\theta = z + \frac{1}{z}$$

$$\Rightarrow \cos\theta = \frac{1}{2} \left(z + \frac{1}{z} \right) = \frac{z^2 + 1}{2z}$$

$$\therefore \cos\theta = \frac{z^2 + 1}{2z} = \frac{z(z + \frac{1}{z})}{2z} = \frac{z + \frac{1}{z}}{2}$$

$$\text{Also, } z = e^{i\theta}$$

$$\Rightarrow dz = i e^{i\theta} d\theta$$

$$\Rightarrow d\theta = \frac{dz}{i e^{i\theta}} = \frac{dz}{iz}$$

From (1) we get,

$$I = \int_0^{2\pi} \frac{d\theta}{5+2\cos\theta} = \oint \frac{\frac{dz}{iz}}{5+2 \cdot \frac{z^2+1}{2z}} \quad \left[\text{Where, } z: |z|=1 \right]$$

$$\Rightarrow I = \int_0^{2\pi} \frac{\frac{dz}{iz}}{5+z+\frac{1}{z}} = \frac{1}{i} \int_0^{2\pi} \frac{1}{5z+z^2+1} dz = \frac{1}{i} \oint f(z) dz \quad \text{--- (4)}$$

$$\left[\text{Where, } f(z) = \frac{1}{5z+z^2+1} \right]$$

Now, the poles of $f(z)$ are obtained by solving

$$5z + z^2 + 1 = 0$$

$$\Rightarrow z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{-5 \pm \sqrt{5^2 - 4 \times 1 \times 1}}{2 \times 1}$$

$$\therefore z = \frac{-5 + \sqrt{21}}{2}, \frac{-5 - \sqrt{21}}{2}$$

Which are simple poles but, only the pole $z = \frac{-5 + \sqrt{21}}{2}$ lies inside

The contour C ,

$$\therefore \text{Res} \left[f(z), \frac{-5 + \sqrt{21}}{2} \right] = \lim_{z \rightarrow \frac{-5 + \sqrt{21}}{2}} \left\{ z - \left(\frac{-5 + \sqrt{21}}{2} \right) \cdot \frac{1}{5z + z^2 + 1} \right\}$$

$$= \lim_{z \rightarrow \frac{-5 + \sqrt{21}}{2}} \left\{ \frac{z^2 + 5 - \sqrt{21}}{2} \cdot \frac{1}{5z + z^2 + 1} \right\}$$

$$= \lim_{z \rightarrow \frac{-5 + \sqrt{21}}{2}} \left\{ \frac{z^2 + 5 - \sqrt{21}}{2} \cdot \frac{1}{\left(z - \frac{-5 + \sqrt{21}}{2} \right) \left(z - \frac{-5 - \sqrt{21}}{2} \right)} \right\}$$

$$= \lim_{z \rightarrow \frac{-5 + \sqrt{21}}{2}} \left\{ \frac{\frac{z^2 + 5 - \sqrt{21}}{2}}{\frac{(2z + 5 - \sqrt{21})(2z + 5 + \sqrt{21})}{4}} \right\}$$

$$= \lim_{z \rightarrow \frac{-5 + \sqrt{21}}{2}} \left\{ \frac{2(z^2 + 5 - \sqrt{21})}{(2z + 5 - \sqrt{21})(2z + 5 + \sqrt{21})} \right\}$$

$$= \lim_{z \rightarrow \frac{-5 + \sqrt{21}}{2}} \left\{ \frac{2}{2z + 5 + \sqrt{21}} \right\}$$

$$= \frac{1}{\sqrt{21}} \quad \underline{\underline{\text{Ans}}}$$